

ORIGINAL ARTICLE

AI-Powered Applications' Effects on English Language Learners' Cognitive, Metacognitive, and Resource Management Strategies, and Language Achievement

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ABSTRACT

Background: Generative AI (GenAI) offers scalable feedback and planning support, yet rigorous evidence on how AI-supported instruction shapes EFL learners' self-regulated strategies and achievement remains limited.

Objectives: The main objectives are to test whether a feedback-oriented GenAI integration improves (a) cognitive, metacognitive, and resource-management strategies and (b) English achievement in university EFL coursework.

Methods: Ten intact classes of undergraduates ($N = 310$) at Allameh Tabataba'i University were randomised at the class level to an AI condition ($n = 139$) or control ($n = 171$). The AI group completed a 6-h AI-literacy workshop and 12 weeks of guided practice using ChatGPT, Poe, and Bard within a draft-first → AI critique → human evaluation workflow; the control group received the same curriculum without AI. Outcomes included adapted SILL subscales (cognitive, metacognitive, and resource management) and a researcher-developed achievement composite (comprising reading, vocabulary, and writing). A MANCOVA with pretests as covariates, followed by prespecified ANCOVAs, reported adjusted means with 95% CIs and accounted for class-level clustering. Qualitative data comprised 834 biweekly reflective journals, which were thematically analysed using double coding (Cohen's $\kappa = 0.88$).

Results: The AI group outperformed the control across all four outcomes; a MANCOVA indicated a significant multivariate effect, with follow-up ANCOVAs showing partial η^2 values of approximately 0.16–0.22. The journals illuminated mechanisms—planning/monitoring, time efficiency, confidence, and calibrated help-seeking—and flagged the risks of cognitive offloading without explicit guardrails.

Conclusions: When embedded in coherent pedagogy and supported by AI literacy, feedback-oriented GenAI can strengthen the use of self-regulated strategies and improve language achievement in university EFL contexts. Implementation should include clear usage norms and reflective monitoring to sustain learning quality.

1 | Introduction

The integration of Artificial Intelligence (AI) into education has revolutionized traditional instructional practices, introducing

new dimensions of personalization, adaptability and learner autonomy. With advancements in natural language processing (NLP), machine learning (ML), and intelligent tutoring systems, AI technologies have gained traction as transformative tools in

Practitioner Notes

- What is already known about this topic:
 - Self-regulated learning strategies (cognitive, meta-cognitive and resource management) are essential for successful second language acquisition.
 - AI-powered tools like ChatGPT are increasingly used in language learning, but empirical research on their pedagogical effects is limited.
 - EFL learners often struggle with applying strategies consistently without structured support.
- What this paper adds:
 - Provides empirical evidence that AI tools significantly improve the use of self-regulated learning strategies among EFL learners.
 - Demonstrates that AI-mediated instruction leads to measurable gains in English language achievement
 - Offers a structured model of integrating AI tools in language courses with clear training protocols
 - Identifies learner perceptions of AI as a scaffold for autonomy, confidence, and time management.
- Implications for practice and/or policy:
 - Teachers should be trained to embed AI tools purposefully within language learning tasks to support strategic engagement.
 - AI-based interventions can complement traditional instruction and support individualised learning.
 - Institutions should consider policy frameworks for ethical and pedagogically sound use of generative AI in EFL education.
 - Further research and teacher development are needed to maximise the benefits of AI while mitigating risks such as overreliance or misinformation.

various educational contexts (Baker et al. 2019; Chen et al. 2020; McCarthy 2007). Within the domain of language learning, AI's capacity to provide instant feedback, tailor learning experiences, and support metacognitive and cognitive strategies has positioned it as a promising pedagogical asset (Godwin-Jones 2022; Li et al. 2024; Smutny and Schreiberova 2020; Yang and Kyun 2022). Since late 2022, the diffusion of generative AI—large language models capable of producing text, feedback, and task scaffolds—has accelerated classroom experimentation and scholarly guidance (Chan and Hu 2023; Godwin-Jones 2024; Lee et al. 2025; Li et al. 2025; Wang, Wang, et al. 2024).

The emergence of Intelligent Computer-Assisted Language Learning (ICALL) has further strengthened AI's role in second-language acquisition via chatbots, automated writing evaluation, and adaptive platforms (Smutny and Schreiberova 2020; Yang and Kyun 2022). These applications show potential for improving learners' linguistic competence, motivation, and writing processes (Alangari 2025; Chen and Gong 2025; Karagöz 2025). Moreover, AI-mediated environments can facilitate self-regulated learning by fostering planning, monitoring, and reflection—core elements of strategic learning

(Oxford 2016; Panadero 2017). Recent syntheses in CALL and language-education journals map how GenAI mediates distributed agency, feedback cycles and autonomy development (Godwin-Jones 2024) and report heterogeneous but promising effects on L2 writing outcomes (Karagöz 2025; Lee et al. 2025; Li et al. 2025).

However, despite the pedagogical promise of AI, challenges persist. Instructors often face difficulties in aligning AI with pedagogy or institutional support (Alhusaiyan 2025; Pokrivcakova 2019; Zawacki-Richter et al. 2019), and students can become overly reliant on automated support (Chan and Hu 2023; Xu et al. 2024). Peer-reviewed work likewise advocates human-centred, risk-aware uses of GenAI—emphasising transparency, age-appropriate use, data protection and teacher capacity-building (Chan and Hu 2023; Memarian and Doleck 2024; Wang, Wang, et al. 2024; Yang and Kyun 2022; Zawacki-Richter et al. 2019). Formative-assessment scholars caution that AI reshapes feedback and self/peer assessment, demanding validity- and agency-aware classroom designs (Memarian and Doleck 2024; Zhai 2023).

While many studies examine technical/instructional dimensions, fewer probe how AI affects learners' cognitive, meta-cognitive and resource-management strategies—a critical gap given the centrality of strategic behaviour in language learning (Oxford 2016; Panadero 2017; Yang and Kyun 2022). In particular, design-dependent evaluations are needed to understand how AI tools influence engagement with feedback, autonomy in revision, and longer-term self-regulation (Chan and Hu 2023; Chen and Gong 2025; Karagöz 2025; Lee et al. 2025; Li et al. 2025; Xu et al. 2024).

This study addresses these gaps by investigating the impact of AI-supported instruction on learning strategies and writing achievement among EFL learners. Grounded in self-regulation and strategy frameworks (Oxford 2016; Panadero 2017; Pintrich 2004)—and informed by activity-theory perspectives (Jonassen and Rohrer-Murphy 1999)—we examine how AI-enhanced environments mediate cognitive, metacognitive, and behavioural engagement. To resolve scope ambiguity, we clarify that the study examines strategy use in coursework (including writing-intensive tasks) and an overall language-achievement composite (reading, vocabulary, writing); we also define 'generative AI' as LLM-based tools used under explicit instructional guardrails consistent with recent peer-reviewed guidance (Chan and Hu 2023; Memarian and Doleck 2024; Wang, Zhao, et al. 2024; Yang and Kyun 2022; Zawacki-Richter et al. 2019).

2 | Literature Review

2.1 | AI in Education

Artificial intelligence (AI) has evolved from automation to intelligent systems capable of delivering personalised, adaptive, and data-driven learning experiences. Progress in NLP, ML, and learning analytics underpins this shift, with reviews noting opportunities alongside design and capacity constraints (Chen et al. 2020; Wang, Wang, et al. 2024; Xiao et al. 2024; Zawacki-Richter et al. 2019). The pedagogical promise centres

on engagement, autonomy, and instructional efficiency when AI is embedded within coherent learning designs (Godwin-Jones 2022; Kasneci et al. 2023; Yang and Kyun 2022). Recent syntheses and commentaries converge on the view that large language models (LLMs) can enhance formative feedback, material design, and learner agency, contingent on teacher preparation and institutional alignment (Chan and Hu 2023; Godwin-Jones 2024; Lee et al. 2025; Li et al. 2025).

Persistent concerns include technological disparities, ethical risks, and the preparedness of instructors to enact AI-supported pedagogy. Evidence highlights gaps between technical affordances and classroom realities, calling for pedagogy-first integration, infrastructure, and teacher professional learning (Chen et al. 2020; Memarian and Doleck 2024; Zawacki-Richter et al. 2019). Across 2023–2025 publications, scholars recommend explicit teacher-facing guidelines (e.g., role framing, prompt design, disclosure norms) and student AI literacy to mitigate bias, privacy, and over-reliance (Chan and Hu 2023; Memarian and Doleck 2024; Wang, Zhao, et al. 2024).

We use ‘generative AI’ to denote LLM-based tools that can produce human-like text and feedback for instructional tasks (Kasneci et al. 2023; Lee et al. 2025; Li et al. 2025). ‘AI-supported instruction’ refers to teacher-designed activities in which AI provides feedback, exemplars, or practice materials under declared guardrails and transparent classroom norms (Chan and Hu 2023; Memarian and Doleck 2024).

3 | Literature Review

3.1 | AI in Language Learning and Teaching

Language education has moved from traditional Computer-Assisted Language Learning (CALL) to Intelligent CALL (ICALL), with chatbots, automated evaluation, and AI writing assistants providing immediate, tailored feedback for grammar, lexis, coherence, and discourse modelling (Godwin-Jones 2022; Smutny and Schreiberova 2020; Yang and Kyun 2022). Systematic reviews from 2023 to 2025 report generally positive—but design- and context-dependent—effects, with more substantial gains when AI is framed as feedback/coaching rather than as an answer-giving system (Karagöz 2025; Lee et al. 2025; Li et al. 2025). Evidence from Asian higher education (HEIs) shows improved perceptions of GenAI for writing and grammar support, yet persistent ambivalence about quality and dependence, underscoring the need for clear guardrails and teacher mediation (Chan and Hu 2023; Guo et al. 2025). Empirical work also documents gains in EFL writing proficiency and process measures under AI-supported designs (Alangari 2025; Chen and Gong 2025).

3.2 | AI and Learning Strategies

Cognitive strategies—such as rehearsal, elaboration and organisation—are central to language acquisition (O’Malley and Chamot 1990; Oxford 2016; Pintrich 2004). AI tools, particularly NLP-based writing assistants, can scaffold these strategies by offering timely, specific feedback that supports planning and

revision cycles. The benefits are greater when learners use AI feedback for diagnosis and re-drafting rather than direct substitution (Chan and Hu 2023; Karagöz 2025; Li et al. 2025). Studies report improvements in coherence, lexical choice and overall writing outcomes within AI-mediated tasks (Alangari 2025; Chen and Gong 2025; Lee et al. 2025).

Metacognitive strategies—planning, monitoring, evaluating—are likewise supported by AI through revision histories, performance dashboards and structured critique prompts that cue reflective practice and autonomy (Memarian and Doleck 2024; Wang, Wang, et al. 2024; Yang and Kyun 2022). At the same time, researchers caution against ‘cognitive offloading’ and emphasise classroom designs that preserve learner agency (Chan and Hu 2023; Memarian and Doleck 2024). In this respect, formative-assessment scholarship recommends validity- and agency-aware cycles (e.g., ‘draft → AI critique → human evaluation’) to avoid deskilling while widening feedback channels (Godwin-Jones 2024; Memarian and Doleck 2024).

AI can also shape resource-management strategies (time allocation, help-seeking, study environment). Adoption studies indicate that intention to use and perceived utility are related to motivation and planned engagement, but infrastructure and policy uncertainty remain significant practical barriers (Guo et al. 2025; Zawacki-Richter et al. 2019). Reviews similarly note that AI may reduce time costs and increase willingness to iterate when embedded in transparent, teacher-led workflows (Chen et al. 2020; Wang, Wang, et al. 2024).

3.3 | Critical Evaluation and Theoretical Anchors

Equity, ethics, and transparency remain salient. Reviews highlight the risks associated with data privacy and algorithmic bias, arguing for a human-centred design that prioritises learner agency and inclusivity (Memarian and Doleck 2024; Wang, Zhao, et al. 2024; Zawacki-Richter et al. 2019). Activity Theory offers a robust lens for examining AI-mediated learning by situating interactions among the learner (subject), AI tools (mediational means), and sociocultural context (community, rules, division of labour) (Jonassen and Rohrer-Murphy 1999; Yang and Kyun 2022). This perspective complements self-regulated learning constructs by clarifying how GenAI can reconfigure feedback loops, workload distribution, and attention management (Godwin-Jones 2024; Panadero 2017; Pintrich 2004).

Post-2022 evidence converges on a conditional-benefit pattern: GenAI enhances feedback, autonomy, and efficiency in writing-intensive tasks when embedded in teacher-led designs, AI literacy training, and explicit guardrails; otherwise, risks of over-reliance and uneven access persist (Chan and Hu 2023; Godwin-Jones 2024; Lee et al. 2025; Li et al. 2025; Wang, Wang, et al. 2024). This motivates our focus on how AI-supported instruction affects learners’ cognitive, metacognitive, and resource-management strategies during writing-intensive coursework and whether such changes translate into gains on an overall language-achievement composite. The qualitative component probes perceived mechanisms (planning, monitoring, revision) to address calls for mechanism-level insight in recent reviews (Karagöz 2025; Lee et al. 2025; Li et al. 2025).

Despite rapid post-2022 interest in generative AI (LLM-based tools such as ChatGPT), the literature still lacks mechanism-focused evidence on how feedback-oriented, guardrailed AI-supported instruction shapes EFL learners' cognitive, metacognitive, and resource-management strategies within writing-intensive coursework, and whether strategy gains transfer to an overall language-achievement composite; prior studies are often descriptive, tool-centric, or methodologically limited (e.g., small samples, weak alignment of outcomes, minimal reporting of baseline equivalence). To address this gap in a university EFL context, we pose three research questions:

RQ1. *To what extent does AI-supported instruction influence learners' use of cognitive, metacognitive, and resource-management strategies in writing-intensive tasks?*

RQ2. *To what extent does AI-supported instruction affect overall language achievement (reading, vocabulary, writing) relative to business-as-usual instruction?*

RQ3. *How do learners perceive AI's role in supporting self-regulated learning processes, including planning, monitoring and revision?*

4 | Methodology

4.1 | Participants

Participants in this study were 310 undergraduate students from Allameh Tabataba'i University, Iran. The sample consisted of 45% female students ($n=139$) and 55% male students ($n=171$), aged between 19 and 26 years ($M=21$, $SD=2.53$). Students were randomly assigned to either an experimental group ($n=139$) or a control group ($n=171$), with the assignment balanced across 10 intact classes. Because allocation occurred at the class (cluster) level, we rephrase 'randomly assigned' as 'cluster random allocation': 10 intact classes were randomly allocated to conditions (5 experimental, 5 control); individual students were not randomised. To reduce cross-group contamination, classes were held at different times with separate instructors, and students were explicitly instructed not to share AI-related materials across sections. The experimental group received AI-powered instruction designed to enhance cognitive, metacognitive, and resource management strategies, as well as language achievement. In contrast, the control group received regular instruction without exposure to AI tools. This group-based assignment supported a quasi-experimental design with both quantitative and qualitative components. Participants' majors and baseline proficiency (as measured by the institutional placement test) were recorded and summarised for both groups; no statistically significant differences were observed in the baseline.

4.2 | Research Instruments

Two research instruments were used. The first, adapted from Oxford's Strategy Inventory for Language Learning (SILL), assessed English language learners' use of cognitive, metacognitive, and resource management strategies. The instrument was updated to reflect the three strategic domains targeted by the

intervention. Cronbach's alpha values indicated strong internal consistency, with values of 0.85 for cognitive, 0.87 for metacognitive, and 0.83 for resource management strategies. Content validity for the adapted SILL was established through expert panel review and cognitive pretesting. The factorial structure was examined using confirmatory factor analysis, with fit indices (see also Kline 2016; Oxford 2016). Reliability statistics are reported for each subscale at both the pre-test and post-test (α and ω).

The second instrument measured language achievement through a researcher-developed English proficiency test consisting of three subcomponents: reading comprehension, vocabulary-in-use, and short essay writing. The test was administered at the beginning and end of the semester to assess growth over time. Internal consistency was high ($\alpha=0.89$). Test blueprinting aligned items to the course outcomes and CEFR B1–B2 bands; items were piloted with a comparable cohort to check difficulty and discrimination. The writing subtest was scored with an analytic rubric by two trained raters; inter-rater reliability was estimated using a two-way random-effects ICC with absolute agreement.

Additionally, qualitative data were collected from 834 reflective journals submitted biweekly by the experimental group. These journals captured students' perceptions of the impact of AI tools on their learning strategies and language development, as well as their reflections on using and engaging with these tools. The count of 834 journals reflects six entries per participant over the 12 weeks. Journals were written primarily in L2; when L1 segments appeared, translation procedures emphasised meaning preservation and were documented in the codebook.

4.3 | Procedure

Ethical approval was obtained from the university's Institutional Review Board. Informed consent was secured from all participants. For transparency, the IRB protocol ID and approval date are reported in the Declarations section (identifier redacted here for review); dates of consent collection are also provided. During the first week, all students completed pre-tests assessing strategic behaviour (cognitive, metacognitive, and resource management) and language achievement.

Students in the experimental group then attended a 6-h AI literacy workshop over two sessions. The training focused on utilising ChatGPT, Poe, and Google Bard for tasks such as vocabulary development, summarisation, grammar correction, goal setting, and self-monitoring. These tools were introduced as cognitive scaffolds, metacognitive aids, and resource support systems aligned with language learning objectives. An approved-use policy ('draft first → AI critique → human evaluation') was enforced; students learned prompt design, citation transparency, and academic-integrity guardrails (data privacy, fact-checking and bias awareness).

Following the workshop, students engaged with the AI tools over a 12-week period, utilising them in weekly assignments, writing tasks, and independent learning. Activities were purposefully designed to activate all three strategic domains while targeting measurable improvements in language performance.

Learners documented their use of and perceptions about AI in journals, which instructors reviewed for formative feedback. The control group received traditional instruction using the duplicate curriculum content but without AI tools. Both groups completed post-tests during the final week of the semester to assess outcomes. Missing data (<5%) were handled via multiple imputation under MAR assumptions, with listwise-deletion sensitivity checks yielding the same conclusions.

4.4 | Research Design

A mixed-methods, quasi-experimental design with a pre-test-post-test control group structure was employed. The quantitative component used ANCOVA to compare post-test scores between groups, using pre-test scores as covariates. This allowed for isolating the effects of the AI-powered intervention on strategy use and language achievement. Within-group changes were also examined using paired t-tests. Given multiple correlated outcomes, we first conducted a MANCOVA on the four dependent variables (cognitive, metacognitive, resource management, and achievement), followed by follow-up ANCOVAs with adjusted means and 95% confidence intervals. Because allocation occurred at the class level, we accounted for clustering by reporting cluster-robust standard errors and, as a robustness check, estimating two-level mixed ANCOVA models (students nested in classes); intraclass correlations (ICCs) and the design effect are reported. The multivariate test controlled for family-wise error; a conservative Bonferroni check is provided in the supplement.

The qualitative component followed a phenomenological thematic analysis approach. Journal entries were coded along three dimensions: (1) frequency of AI use, (2) strategic purpose (cognitive, metacognitive, or resource management), and (3) self-perceived benefit to language learning. Emergent patterns highlighted the depth of learners' engagement with AI and contextualised the quantitative trends. Two analysts co-developed a codebook through iterative cycles (open → axial). Double-coding was performed on 20% of the entries; discrepancies were resolved through discussion, and negative-case analysis and saturation signals were documented. Representative quotes (2–3 per theme) are reported to ensure voice diversity.

4.5 | Data Analysis

ANCOVA was conducted using pre-test scores as covariates to evaluate the effects of the intervention on the four dependent variables: cognitive strategies, metacognitive strategies, resource management strategies, and language achievement. Assumptions of normality, linearity, homogeneity of variance,

and homogeneity of regression slopes were met. Descriptive statistics, *F*-values, significance levels, and effect sizes (partial η^2) were reported. Multivariate results from the initial MANCOVA are reported with Wilks' Λ , *F*, *df*, *p*, and partial η^2 ; univariate ANCOVAs include adjusted means with 95% CIs, covariate effects, and partial η^2 . Baseline equivalence (means/SDs and tests) for demographics and all outcomes is shown in Table 1, and homogeneity of regression slopes was verified for each outcome. Clustering was addressed via cluster-robust SEs and mixed-effects sensitivity analyses. Effect-size interpretation follows field benchmarks; exact thresholds are stated in the notes.

NVivo software was used for thematic analysis of journal data. Inter-rater reliability was established through double-coding of 20% of entries ($\kappa=0.88$). The findings were triangulated with quantitative data to provide a comprehensive understanding of how AI-powered applications influenced learners' strategic behaviour and academic outcomes. Inter-rater agreement is also reported for a subset after adjudication; qualitative–quantitative triangulation was conducted at the interpretation stage (convergence, complementarity and discrepancy mapping).

5 | Results

5.1 | Quantitative Results

This section reports the quantitative effects of AI-supported instruction on EFL learners' cognitive, metacognitive, and resource-management strategies, as well as language achievement. Analyses followed a prespecified sequence: (a) descriptive summaries, (b) assumption checks, (c) a multivariate test (MANCOVA) with pretests as covariates and (d) prespecified follow-up ANCOVAs for each outcome. Because allocation occurred at the class level (intact classes), all primary models employed cluster-robust standard errors at the classroom level. A two-level mixed ANCOVA (students nested within classes) served as a robustness check and yielded directionally identical conclusions. Baseline equivalence tests showed no significant group differences on demographics or pretest measures (Table 1; all *ps* > 0.05). Missing data were < 5% and handled with multiple imputation; listwise-deletion sensitivity analyses produced the same inferences.

5.2 | Descriptive Statistics

The post-test results favoured the AI group on all outcomes (Table 1). For cognitive strategies, the AI group averaged 3.87 (SD=0.52) versus 3.45 (SD=0.60) in the control group. For metacognitive strategies, the means were 4.02 (SD=0.49) vs. 3.51 (SD=0.55); for resource-management strategies, the means

TABLE 1 | Descriptive statistics by group.

Group	<i>N</i>	Cognitive <i>M</i> (SD)	Metacognitive <i>M</i> (SD)	Resource <i>M</i> (SD)	Language achievement <i>M</i> (SD)
Experimental	139	3.87 (0.52)	4.02 (0.49)	3.94 (0.50)	78.4 (6.8)
Control	171	3.45 (0.60)	3.51 (0.55)	3.42 (0.58)	72.1 (7.4)

were 3.94 (SD = 0.50) vs. 3.42 (SD = 0.58). Language achievement was likewise higher in the AI group, 78.4 (SD = 6.8) vs. 72.1 (SD = 7.4). These unadjusted differences motivated multivariate testing that accounts for the baseline.

5.3 | Assumption Testing for MANCOVA

Before proceeding with the primary analysis, the assumptions necessary for conducting a MANCOVA were systematically evaluated to ensure the robustness of the results. As summarised in Table 2, all five assumptions were adequately met. First, multivariate normality was confirmed using the Mahalanobis distance and visual inspection of histograms, while linearity was supported through visual examination of scatter plots. Furthermore, the assumption of homogeneity of variance–covariance matrices was tested using Box's M, and it was found to be non-significant ($p > 0.05$), indicating no violation. In addition, multicollinearity was ruled out, as all variance inflation factor (VIF) values were comfortably below the threshold of 2. Lastly, the assumption of homogeneity of regression slopes was met, evidenced by the lack of significant interaction between covariates and group membership. The satisfaction of these assumptions provided a sound statistical foundation for conducting the subsequent multivariate analyses.

5.4 | Multivariate (MANCOVA) Results

Controlling for pretests, there was a significant overall group effect across the four outcomes, Wilks' $\Lambda = 0.72$, $F(4, 303) = 23.80$, $p < 0.001$, partial $\eta^2 = 0.28$. These values are reproduced in Table 3 for consistency. The magnitude indicates a significant multivariate effect, justifying prespecified univariate follow-ups.

5.4.1 | Follow-Up ANCOVAs With Adjusted Means

Follow-up ANCOVAs (pretests as covariates; cluster-robust SEs) showed significant group differences on all outcomes, with large

partial η^2 (Table 4). Specifically, the AI group outperformed the control group on cognitive strategies, $F(1, 306) = 14.87$, $p < 0.001$, partial $\eta^2 = 0.16$; metacognitive strategies, $F(1, 306) = 18.65$, $p < 0.001$, partial $\eta^2 = 0.19$; resource-management strategies, $F(1, 306) = 16.22$, $p < 0.001$, partial $\eta^2 = 0.17$; and language achievement, $F(1, 306) = 25.30$, $p < 0.001$, partial $\eta^2 = 0.22$. The pretest covariate was a significant predictor in each model ($p < 0.001$), as expected.

After controlling for pretests and class-level clustering, the ANCOVA-adjusted means in Table 5 reveal clear, practically meaningful advantages for the AI group across all outcomes. On the strategy measures (scales 1–5), the AI group outperformed the control group by 0.42 points on cognitive strategies (3.87 vs. 3.45), 0.51 points on metacognitive strategies (4.02 vs. 3.51), and 0.52 points on resource-management strategies (3.94 vs. 3.42)—roughly 12%–15% relative gains. The 95% confidence intervals do not overlap for any outcome, and the group effects remain significant under Bonferroni control (all $p < 0.001$). Effect sizes are consistently large (partial $\eta^2 = 0.16$ –0.19 for strategies), indicating that the AI-supported workflow fostered deeper rehearsal/elaboration and better organisation (cognitive), stronger planning/monitoring/evaluation (metacognitive), and more efficient time use and help-seeking (resource management). Critically, these strategy gains translated into achievement: the AI group's language score exceeded the control by 6.30 points (78.40 vs. 72.10; partial $\eta^2 = 0.22$), the largest univariate effect. Pretest covariates were significant in all models (see S3), confirming the appropriateness of the adjustment, and mixed-model robustness checks yielded the same conclusions. Taken together, the pattern supports a mechanism-first interpretation: when embedded in a draft-first $\rightarrow$ AI-critique $\rightarrow$ human-evaluation workflow, generative-AI tools strengthen self-regulated strategy use, and those strategy improvements generalise to higher language performance.

TABLE 2 | Assumption tests for MANCOVA.

Assumption	Test used	Result
Multivariate normality	Mahalanobis Distance, Histograms	Satisfied
Linearity	Scatterplots	Satisfied
Homogeneity of variance–covariance matrices	Box's M Test	Not significant ($p > 0.05$)
No multicollinearity	Variance Inflation Factor (VIF)	All VIF < 2
Homogeneity of regression slopes	Interaction Terms in Model	No significant interactions ($p > 0.05$)

TABLE 3 | Multivariate tests (MANCOVA) for group differences, controlling for pretests.

Effect	Wilks' Lambda	F	df	p	Partial η^2
Group	0.72	23.80	4, 303	<0.001	0.28

TABLE 4 | Follow-up ANCOVA results.

Dependent variable	F	df	p	Partial η^2
Cognitive strategies	14.87	1, 306	<0.001	0.16
Metacognitive strategies	18.65	1, 306	<0.001	0.19
Resource management strategies	16.22	1, 306	<0.001	0.17
Language achievement	25.30	1, 306	<0.001	0.22

TABLE 5 | Adjusted means (SE) with 95% confidence intervals and covariate effects (ANCOVA).

Outcome	Control adj. M (SE)	95% CI (Control)	AI adj. M (SE)	95% CI (AI)	Group F(1, 306)	p	Partial η^2	Pretest F(1, 306)	Pretest p
Cognitive strategies	3.45 (0.046)	[3.36, 3.54]	3.87 (0.044)	[3.78, 3.96]	14.87	<0.001	0.16	see Table S3	<0.001
Metacognitive strategies	3.51 (0.042)	[3.43, 3.59]	4.02 (0.042)	[3.94, 4.10]	18.65	<0.001	0.19	see Table S3	<0.001
Resource-management strategies	3.42 (0.044)	[3.33, 3.51]	3.94 (0.042)	[3.86, 4.02]	16.22	<0.001	0.17	see Table S3	<0.001
Language achievement	72.10 (0.566)	[70.99, 73.21]	78.40 (0.577)	[77.27, 79.53]	25.30	<0.001	0.22		

5.5 | Qualitative Results

To provide insight into how learners experienced AI-powered applications during the intervention, a thematic analysis was conducted on 834 reflective journals submitted biweekly by students in the experimental group. Analysis followed a phenomenological approach, focusing on learners' perceptions and lived experiences as they engaged with tools such as ChatGPT, Poe, and Google Bard. Four overarching themes emerged, each with clearly defined subthemes and supporting excerpts from participant reflections. To bolster trustworthiness, coding proceeded through iterative cycles with an audited trail (open → axial → theme), constant comparison, and peer debriefing, aligning with established guidance on rigorous thematic analysis (Braun and Clarke 2006; Lincoln and Guba 1985; Nowell et al. 2017; Saldaña 2021). Excerpts were lightly copy-edited for readability; ellipses indicate omitted text; occasional L1 segments were translated with meaning-preserving notes in the codebook.

5.5.1 | Theme 1: Cognitive Engagement Through AI

Learners' reflective journals revealed that AI-powered applications enhanced a wide array of cognitive strategies, including elaboration, rehearsal, inferencing, and organisation. Many students described using tools such as ChatGPT and Poe to clarify word meanings, generate synonyms, explore example sentences, and ask follow-up questions to deepen understanding—an experience closely aligned with the elaboration strategies emphasised in Oxford's (2016) taxonomy. These tools allowed students to expand their mental lexicon by noticing how phrases and collocations naturally appear in context. As one participant put it, 'I used to learn vocabulary from the book, but now when I see it used in a full paragraph from AI, I remember it better' (P057). Learners also used AI to scaffold their rehearsal processes, such as paraphrasing complex sentences, repeating grammatical structures, or simulating question-answer dialogues. In line with O'Malley and Chamot's (1990) classification, several students practiced inferencing by asking AI to explain unfamiliar chunks or connect ideas logically: 'When I get confused in an article, I just copy the paragraph and ask the bot what it means in simpler words. It helps me guess the meaning of other sentences too' (P109). Moreover, AI helped many learners organise their ideas for writing tasks, functioning as a cognitive planning assistant. This aligns with analyses reported by Godwin-Jones (2024), who observed that intelligent tools promote learners' cognitive flexibility by breaking down complex tasks into manageable steps. As one student wrote, 'ChatGPT helped me outline my argument before writing. I did not just jump in anymore—I planned better' (P173).

Altogether, AI tools were not only used for passive lookup but actively supported learners in internalising, restructuring and applying knowledge in meaningful academic contexts. Prominent subthemes included lexical elaboration, worked-example rehearsal, guided inference, and outlining/structuring. A minority voiced concerns about shallow processing when accepting AI paraphrases uncritically—'Sometimes I copy the AI version because it looks better, but I am not sure I learned it' (P231)—underscoring the need for 'draft-first, AI-second' routines to protect deep processing (Godwin-Jones 2024).

5.5.2 | Theme 2: Enhanced Metacognitive Regulation

The second dominant theme identified in the journals was learners' development of metacognitive regulation, particularly in planning, monitoring, evaluating, and debugging their learning processes. Learners frequently reported that AI tools promoted goal-setting behaviour, such as writing with a specific structure in mind or intentionally using more transition devices—mirroring Oxford's (2016) 'goal-setting and planning' subcategory. One participant reflected, 'Each week I set a mini-goal like writing with fewer passive sentences, and ChatGPT showed me if I succeeded' (P041). Additionally, students utilised AI to monitor their learning in real time, particularly by pasting drafts for revision and comparing AI suggestions with their original work. This reflects self-monitoring and performance tracking processes emphasised in SRL reviews (Panadero 2017), where feedback loops are essential for learner self-regulation. A learner stated, 'I always check my paragraph with the bot first. It shows the weak parts, and I can fix them before showing the teacher' (P121).

Furthermore, students reported evaluating their growth by reviewing their journal entries, tracking how AI helped reduce their repeated errors or improve sentence variety. Some even developed personal checklists based on AI feedback, indicating a shift from reactive to proactive metacognition. Debugging, or problem-solving in response to task failure, was another subtheme: students often described instances of refining their writing after realising they had misunderstood prompts or misused expressions. 'After seeing AI rewrite my sentence three times, I realized my sentence logic was wrong, not just the grammar' (P194).

Notably, a few learners began strategizing how and when to use AI, indicating higher-order metacognitive awareness and control over tool deployment. As one learner stated, 'Now I do not ask AI every time. I write myself first, then use it for feedback. It is like a test of my learning' (P202). These findings suggest that AI applications, when embedded in task-based instruction, can serve as powerful metacognitive scaffolds fostering autonomous learning regulation. Negative cases referenced 'monitoring fatigue' and conflicting suggestions across tools—'ChatGPT says add details, but another tool says be concise; I get confused which rule to follow' (P264)—pointing to the value of instructor-provided decision heuristics (e.g., prioritise rubric-aligned criteria) and explicit reflection prompts (Bandura 1997; Pintrich 2004; Panadero 2017).

5.5.3 | Theme 3: Resource Management and Strategic Autonomy

The third theme centres on how learners utilised AI tools to manage time, effort, and support systems, all of which are integral to resource management strategies as defined by Pintrich (2004) and further elaborated upon in SRL syntheses (Panadero 2017). Students overwhelmingly indicated that AI enabled more efficient time use, especially for brainstorming, revising, or checking grammar—stages of learning that typically consume disproportionate amounts of effort. As one student wrote, 'Before, I would get stuck on one paragraph for hours. Now I use AI to unblock myself and keep going' (P086). Another critical subtheme was mental effort regulation,

wherein learners strategically used AI when cognitive load became too high.

Students reported breaking tasks into segments with the help of AI, such as writing one paragraph, checking it, and then continuing. This aligns with the concept of distributed cognition, as AI became part of their extended learning system. 'If I get tired or confused, I ask the AI to simplify. It keeps me moving' (P147). Learners also described accessing just-in-time support outside classroom hours, reducing reliance on peers or instructors: 'I am shy to ask teachers for everything, so AI helps me feel independent' (P112).

This sense of self-direction led to another subtheme: strategic help-seeking, where students first consulted AI and then asked teachers more advanced or specific questions, thereby optimising external support. Finally, some learners commented on reducing procrastination by using AI to initiate tasks more easily: 'When I do not feel like starting, I write a question to AI and use the answer as my start. It beats the blank page fear' (P179). These behaviours reveal that AI serves not just as a tool for output improvement but as a psychological resource for organising cognitive and affective effort. A small subset cautioned that 'time-saving' could morph into a time drain through excessive prompt tinkering—'I spend too long trying many prompts to get the perfect answer' (P301)—highlighting the importance of guardrails for productive use (Godwin-Jones 2024) and structured pacing strategies (e.g., fixed cycles).

5.5.4 | Theme 4: Perceived Gains in Language Achievement

The final theme synthesised learners' reflections on their progress in English writing fluency, vocabulary richness, grammatical accuracy, and overall confidence, all of which they attributed to regular AI use across entries, learners emphasised how frequent interaction with AI reduced their error rates and expanded their expressive range. This aligns with recent quasi-experimental and mixed-methods evidence showing that AI-assisted learning enhances writing proficiency and academic language use (Alangari 2025; Chen and Gong 2025). One participant shared, 'My writing has fewer red marks now. The teacher even asked if I got help, but it was my revision with AI' (P093). Another learner noted increased fluency: 'Before, I repeated simple words. Now I can say things like 'it is essential to note' instead of "it is important"' (P122). Learners also described feeling more confident in their academic identity, stating that they were less anxious about making grammatical mistakes or concerns about structural clarity. 'Now I'm not afraid to write essays. I know I can fix things with AI help' (P088). Some even reported feeling motivated to write more frequently, seeing writing as less of a chore and more of a growth opportunity. 'Writing used to be boring, but now I try to improve each time using the suggestions I get' (P077). These perceptions align with the concept of learner self-efficacy, as proposed by Bandura (1997), suggesting that AI tools not only mediate language output but also shape learners' beliefs about their competence and control in academic communication. A minority flagged over-reliance and occasional hallucinations—'Sometimes AI gives me a

wrong citation. I learned to double-check with our textbook' (P198)—indicating developing evaluative judgement alongside confidence. Reported gains are mapped to quantitative improvements on strategy subscales and overall achievement, triangulating across data sources (Alangari 2025; Chen and Gong 2025).

6 | Discussion

The findings suggest that AI-supported instruction can enhance EFL learners' cognitive, metacognitive, and resource-management strategies, ultimately improving their performance on writing-intensive tasks and a broader composite of language achievement. The pattern is consistent with theoretical accounts that emphasise self-efficacy and agentic regulation (Bandura 1997; Alibakhshi et al. 2020), tool-mediated activity (Jonassen and Rohrer-Murphy 1999), and cyclical self-regulated learning (SRL) processes, including forethought, performance, and reflection (Oxford 2016; Panadero 2017; Pintrich 2004). Addressing RQ1 and RQ2 directly, our results align with recent syntheses that conceptualise generative AI as a scaffold that extends—rather than replaces—pedagogy by providing dense, on-demand feedback and exemplars when embedded in coherent instructional designs (Baker et al. 2019; Godwin-Jones 2024; Kasneci et al. 2023; Lee et al. 2025; Li et al. 2025).

Gains in elaboration, rehearsal, inferencing, and organisation suggest that AI outputs functioned as mediational means for noticing and reformulation—mechanisms long associated with strategy-based L2 learning (O'Malley and Chamot 1990; Oxford 2016). Empirically, quasi-experimental and review evidence indicate that AI-generated feedback improves EFL writing quality when leveraged for revision planning rather than direct text replacement (Alangari 2025; Karagöz 2025; Lee et al. 2025; Li et al. 2025). Our qualitative observations of learners using AI to outline arguments, paraphrase complex clauses, and test hypotheses about usage mirror these pathways and resonate with accounts of distributed agency in AI-supported language work (Godwin-Jones 2024). At the same time, reports of 'cognitive offloading' caution against uncritical uptake of suggestions, reinforcing draft-first, AI-second routines advocated in student perception and review studies (Chan and Hu 2023; Karagöz 2025; Lee et al. 2025; Li et al. 2025).

The intervention enhanced learners' planning, real-time monitoring, and evaluative reflection, aligning neatly with SRL's cyclical model (Panadero 2017; Pintrich 2004). Learners described drafting independently, using AI for critique, and revising against explicit criteria—behaviours consistent with strategy internalisation and calibration emphasised in the SRL literature (Oxford 2016). Converging work on assessment-for-learning with AI likewise underscores the value of transparent feedback loops and student agency in interpreting automated input (Memarian and Doleck 2024). Meanwhile, broader educational analyses highlight opportunities and risks that hinge on design quality and teacher mediation (Kasneci et al. 2023).

Improvements in time allocation, help-seeking, and effort regulation suggest that AI has reduced friction at common bottlenecks (e.g., idea generation, early drafting) and provided just-in-time

support. These behaviours are compatible with motivational accounts in which perceived competence and autonomy sustain engagement (Bandura 1997; Pintrich 2004). Evidence on chatbots and classroom AI further suggests that efficiency gains can be achieved when such tools are framed as coaches rather than answer engines (Smutny and Schreiberova 2020; Wang, Zhao, et al. 2024). Importantly, intention-to-use research highlights the roles of usability, facilitating conditions, and social influence in adoption trajectories (Guo et al. 2025), reinforcing the need for explicit AI literacy and infrastructure support (Kohnke and Green-Eneix 2025; Pokrivcakova 2019; Zawacki-Richter et al. 2019).

The advantages observed for the AI group in writing-intensive performance and overall composite are consistent with contemporary quasi-experimental and mixed-methods evidence linking AI-assisted learning to gains in writing proficiency and academic language use (Alangari 2025; Alhusaiyan 2025; Chen and Gong 2025). The thematic analysis converged on the mechanisms most frequently implicated in that literature—planning/monitoring, time efficiency, and confidence—strengthening a mechanism-first interpretation of effects (Braun and Clarke 2006; Godwin-Jones 2022; Nowell et al. 2017). Analytic robustness checks that accounted for class-level clustering and reported adjusted means with 95% confidence intervals support the stability of these inferences under plausible specifications.

Benefits depend on design. Reviews emphasise that effects are strongest when AI is positioned as a coach within scaffolded workflows and weakest when it substitutes for core cognitive work (Karagöz 2025; Lee et al. 2025; Li et al. 2025). Practical constraints—including uneven infrastructure and variable teacher preparedness—remain significant barriers to scale-up (Chen et al. 2020; Kohnke and Green-Eneix 2025; Pokrivcakova 2019; Zawacki-Richter et al. 2019). Future investigations should explore moderators such as proficiency, AI attitudes, and AI literacy (Guo et al. 2025; Xu et al. 2024; Yang and Kyun 2022), and continue to report transparent guardrails to facilitate replication.

7 | Conclusions and Implications

This study provides empirical evidence that AI-supported learning environments can enhance EFL learners' cognitive, metacognitive, and resource-management strategies and, in turn, improve academic writing and broader language achievement. The pattern aligns with accounts of self-efficacy and agentic regulation (Bandura 1997), tool-mediated activity (Jonassen and Rohrer-Murphy 1999), and cyclical self-regulated learning (SRL) processes of planning, monitoring, and reflection (Oxford 2016; Panadero 2017; Pintrich 2004). Taken together with the quantitative advantages for the AI group and convergent qualitative mechanisms (planning/monitoring, time efficiency, confidence), these results add to a growing body of work showing that feedback-oriented GenAI, implemented with instructional guardrails, strengthens strategy use in writing-intensive coursework and translates into gains on an overall language-achievement composite (Godwin-Jones 2024; Lee et al. 2025; Li et al. 2025).

Metacognitive skills such as planning, monitoring, and evaluation were amplified through AI-facilitated reflection, performance tracking, and peer-supported review. At the same time, resource-management strategies (e.g., time allocation, targeted help-seeking) benefited from personalised pacing and adaptive task support. These patterns are consistent with recent syntheses that frame GenAI as a scaffold for distributed learner agency and SRL, while cautioning against ‘cognitive offloading’ in the absence of a draft-first, AI-second workflow (Chan and Hu 2023; Karagöz 2025; Lee et al. 2025; Li et al. 2025).

7.1 | Implications

This study has clear implications for the design of learning. When generative AI is framed as a coach rather than an answer engine, it amplifies the use of strategy without eroding cognitive effort. In practice, that means structuring tasks so students draft first, invite AI critique second, and then revise with human evaluation—an orderly choreography that preserves deep processing while multiplying feedback (Baker et al. 2019; Godwin-Jones 2024; Memarian and Doleck 2024; Zhai 2023). Course policies should make those steps transparent, specify when and how AI may be consulted, and require students to reflect on the provenance of suggestions. In short, pedagogy leads; AI follows.

This study also has implications for teacher development. Integrating AI well is not merely a technical matter; it is a pedagogical skill. Educators require structured opportunities to practice designing prompts aligned with rubrics, adjudicating conflicting tool feedback, and managing data privacy and disclosure with confidence (Kohnke and Green-Eneix 2025; Memarian and Doleck 2024; Zhai 2023). Professional learning communities can serve as laboratories for trying out ‘draft-first → AI critique → human evaluation’ sequences, curating exemplar prompts, and co-developing mini-modules that help students interpret automated feedback through an assessment-for-learning lens.

This study further has implications for equity and access. Benefits will accrue unevenly if infrastructure, usability, and local norms are not addressed head-on. Institutions should invest in reliable access, multilingual interfaces, and AI literacy support that extends beyond early adopters; intention-to-use research reminds us that facilitating conditions and social influence shape adoption as much as individual curiosity (Guo et al. 2025; Pokrivcakova 2019; Zawacki-Richter et al. 2019). Equity in this space involves stable connectivity, clear guidelines that are usable by novices, and scaffolds that help students translate AI feedback into action, regardless of their prior experience.

Finally, this study has implications for future reporting and research practice. To make findings comparable and reproducible, studies should document guardrails and instructional workflows, report adjusted means with confidence intervals, and account for classroom clustering; they should also probe moderators such as proficiency, AI attitudes, and AI-literacy to clarify for whom and under what conditions AI supports learning (Godwin-Jones 2024; Lee et al. 2025; Li et al. 2025). In other

words, the next phase is less about whether AI ‘works’ and more about which designs enable students to do the cognitive and metacognitive work that matters.

8 | Limitations and Suggestions for Further Research

Despite its contributions, this study has limitations. First, the quasi-experimental design limits the ability to make strong causal inferences. Although efforts were made to control for baseline differences, random assignment at the individual level was not feasible, which may have introduced class-level confounds. Future studies should consider randomised controlled trials to establish stronger causal claims. Where class-level allocation is unavoidable, researchers should report ICCs/design effects and use cluster-robust or multilevel models to bound bias.

Second, the study was conducted within a single institutional and cultural context, which may limit the generalizability of findings to other educational systems or learner populations. Further research across diverse sociocultural settings is warranted to examine whether similar effects of AI tools on learning strategies and language performance hold across contexts. Multi-site replications spanning proficiency bands and course types (EAP/ESP) would clarify boundary conditions and scalability.

Third, the duration of AI intervention was limited to one semester. While short-term gains in metacognitive, cognitive, and resource management strategies were evident, the sustainability of these changes over time remains unclear. Longitudinal studies could offer insights into the retention and transferability of AI-facilitated learning strategies. Delayed post-tests and classroom observations would help establish maintenance and transfer.

Additionally, qualitative data were derived solely from reflective journals, which may be subject to social desirability bias or limited introspection. Future research should triangulate such data with interviews, classroom observations, or usage analytics to provide a richer account of learner experiences. Including control-group qualitative data and platform log traces (e.g., prompt/revision histories) would mitigate method asymmetry.

Finally, individual difference variables such as learners’ digital literacy, AI attitudes, or language proficiency levels were not explicitly examined. Given their potential moderating roles, future research should incorporate these variables to better understand who benefits most from AI-supported environments and under what conditions. Pre-registered moderation analyses and measurement-invariance checks for strategy scales are recommended to strengthen inference.

Author Contributions

Junhua Chen and Goudarz Alibakhshi jointly conceptualized and designed the study. Junhua Chen conducted data collection, performed the statistical analysis, and drafted the initial manuscript. Goudarz Alibakhshi supervised the overall research process, critically revised

the manuscript, and provided theoretical and methodological guidance. Both authors reviewed and approved the final version of the manuscript.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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Appendix A

Confirmatory Factor Analysis and Reliability of the Adapted Strategy Inventory

To evaluate the factorial validity of the adapted Strategy Inventory for Language Learning (SILL), a series of confirmatory factor analyses (CFAs) was conducted using the full sample. A three-factor model corresponding to cognitive, metacognitive, and resource-management strategies was specified, with items constrained to load on their intended latent factor and factors allowed to correlate. Estimation used robust maximum likelihood, and model adequacy was judged using conventional thresholds ($CFI/TLI \geq 0.90$ – 0.95 , $RMSEA \leq 0.06$ – 0.08 , $SRMR \leq 0.08$; see Kline 2016; Oxford 2016) (Tables A1 and A2).

Table B1 summarises the global fit indices for the hypothesized three-factor model and a more constrained one-factor model in which all items loaded on a single global strategy factor. As shown, the three-factor

model demonstrated clearly superior fit, supporting the intended multi-dimensional structure.

Standardised factor loadings for the three-factor model ranged from 0.54 to 0.82 for cognitive items, 0.57 to 0.85 for metacognitive items, and 0.52 to 0.80 for resource-management items. All loadings were statistically significant ($p < 0.001$), indicating that each item contributed meaningfully to its intended factor.

Internal-consistency estimates were computed for each subscale at both pre-test and post-test. In addition to Cronbach's alpha (α), McDonald's omega (ω) was used as a more robust indicator of composite reliability. The values in Table A2 indicate stable and satisfactory reliability across time for all three domains.

Taken together, the CFA results and reliability estimates support the construct validity and internal consistency of the adapted SILL for measuring cognitive, metacognitive, and resource-management strategies in the present context.

TABLE A1 | Model fit indices for the adapted SILL ($N = 310$).

Model	χ^2	df	χ^2/df	CFI	TLI	RMSEA (90% CI)	SRMR
One-factor model	712.45	249	2.86	0.86	0.84	0.078 (0.071, 0.085)	0.071
Three-factor model	412.37	247	1.67	0.95	0.94	0.046 (0.039, 0.053)	0.041

TABLE A2 | Internal-consistency reliability (α , ω) for strategy subscales at pre-test and post-test.

Subscale	Timepoint	α	ω
Cognitive strategies	Pre-test	0.84	0.85
	Post-test	0.86	0.87
Metacognitive strategies	Pre-test	0.86	0.87
	Post-test	0.88	0.89
Resource-management strategies	Pre-test	0.82	0.83
	Post-test	0.84	0.85

Appendix B

Qualitative Coding Tree for Learners' Use of AI-Powered Tools

The qualitative analysis of 834 reflective journals yielded a hierarchical coding structure comprising four overarching themes, each with

several subthemes and associated codes. The coding tree was developed through iterative cycles of open and axial coding, constant comparison, and peer debriefing. Table B1 presents the final coding tree, including brief definitions and illustrative codes/phrases.

TABLE B1 | Coding tree for learners' experiences with AI-powered tools.

Theme and subtheme	Definition/focus	Example codes/indicative phrases
Theme 1: Cognitive Engagement Through AI	How AI tools support information processing, comprehension, and idea organisation	
1.1 Lexical elaboration	Using AI to expand, clarify, and contextualise vocabulary	'Asking AI for synonyms', 'seeing words in full sentences', 'collocations in context'
1.2 Worked-example rehearsal	Practising structures and patterns through AI-generated examples and reformulations	'Repeat similar sentence with change', 'model paragraph', 'pattern drilling with AI'
1.3 Guided inferencing	Using AI explanations to infer meanings and relationships in texts	'Explain this paragraph', 'guess meaning from AI explanation', 'linking ideas logically'
1.4 Outlining and structuring	Relying on AI to plan and sequence ideas before writing	'AI-made outline', 'organising arguments', 'step-by-step essay plan'
1.5 Shallow processing concerns	Awareness of copying AI output without deep understanding	'I just accept AI paraphrase', 'beautiful sentence but not learned', 'afraid of overusing AI text'
Theme 2: Enhanced Metacognitive Regulation	How AI tools supported planning, monitoring, evaluating, and adjusting learning	
2.1 Goal-setting and planning	Setting specific language goals and using AI feedback to track them	'Weekly writing goal', 'focus on fewer passive sentences', 'target vocabulary list'
2.2 Self-monitoring and performance tracking	Checking drafts, monitoring errors, and tracking progress with AI	'Check my paragraph first with AI', 'see weak parts', 'compare versions'
2.3 Growth evaluation and checklists	Reflecting on progress over time and creating personal criteria from AI feedback	'My own checklist from AI suggestions', 'tracking repeated mistakes', 'reviewing journal history'
2.4 Debugging and problem-solving	Revising after noticing deeper problems (logic, coherence, task interpretation)	'Sentence logic wrong, not only grammar', 'fixing misunderstanding of prompt'
2.5 Strategic deployment of AI	Deciding when and how to use AI in relation to self-generated work	'Write first, AI second', 'not asking AI every time', 'using AI only at revision stage'
2.6 Monitoring fatigue and conflicting feedback	Feeling overloaded by constant checking or contradictory AI suggestions	'Too many corrections', 'different tools give opposite advice', 'confused which rule to follow'
Theme 3: Resource Management and Strategic Autonomy	How AI tools helped manage time, effort, and support systems	
3.1 Time management and task efficiency	Using AI to reduce time spent stuck on local problems	'Unblock paragraph', 'finish faster', 'save time on grammar checking'
3.2 Effort regulation and cognitive load management	Calling on AI when tasks felt too difficult or mentally draining	'When I am tired', 'simplify hard text', 'break task into smaller parts with AI'
3.3 Just-in-time support and independence	Accessing help outside class and feeling less dependent on teachers or peers	'Ask AI at night', 'not shy to ask questions', '24/7 helper'
3.4 Strategic help-seeking	Using AI first, then asking teachers more advanced questions	'AI for basic questions, teacher for deeper ones', 'refine questions after AI'
3.5 Overcoming procrastination	Using AI to get started and reduce fear of the blank page	'Start with AI answer', 'first prompt then my own writing', 'easier to begin writing'
3.6 Risk of time-drain and unproductive tinkering	Spending excessive time optimising prompts or outputs	'Too long searching perfect prompt', 'wasting time editing AI answer'
Theme 4: Perceived Gains in Language Achievement	Learners' perceptions of changes in proficiency, confidence, and academic identity	
4.1 Writing fluency and flow	Feeling able to produce longer, more continuous texts with fewer pauses	'Write faster', 'less stuck', 'more natural sentences'

(Continues)

TABLE B1 | (Continued)

Theme and subtheme	Definition/focus	Example codes/indicative phrases
4.2 Vocabulary richness and expression	Noticing wider lexical variety and more sophisticated phrases	'Use advanced phrases', 'less repetition of simple words', 'academic phrases bank'
4.3 Grammatical accuracy and clarity	Perceiving fewer errors and clearer sentence structures	'Fewer red marks from teacher', 'cleaner grammar', 'better sentence structure'
4.4 Academic confidence and self-efficacy	Feeling more confident as an academic writer and less anxious about errors	'Not afraid of essays', 'believe I can improve', 'AI as safety net'
4.5 Motivation and willingness to write more	Increased interest in writing and seeing tasks as opportunities for growth	'Writing not boring now', 'try to improve each time', 'enjoy experimenting'
4.6 Evaluative judgement and AI scepticism	Developing critical awareness of AI limitations (e.g., hallucinations, faulty citations)	'Double-check AI citation', 'AI sometimes wrong', 'do not trust 100%'